

in-class-code

Packages

```
# general use
library(tidyverse)

# file organization
library(here)

# data visualization
library(naniar)
library(patchwork)
library(gghighlight)
```

Data

```
veg <- read_csv(here("data", "veg.csv"))

metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

Class code

```
# create a new object from veg
gl_implicitNA <- veg |>
  # filter all occurrences where transect_name contains GL
  filter(str_detect(transect_name, "GL") &
```

```

    site == "ncos") |>
# group by year, pool, and species
group_by(year, transect_name, psoc) |>
# calculate mean percent cover
summarize(mean_pc = mean(percent_cover, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# create a new column called year_pool from year and transect name
unite("year_pool", year, transect_name, remove = TRUE) |>
# join with metadata by year_pool
left_join(metadata, by = "year_pool")

# creating new object from veg
gl_explicit0 <- veg |>
# filter all occurrences where transect_name contains GL
filter(str_detect(transect_name, "GL") &
    site == "ncos") |>
# complete all possible combinations of year, pool, psoc and distance
complete(year, transect_name, psoc, transect_distance,
    # fill values of percent cover with 0
    fill = list(percent_cover = 0)
) |>
# group by year, pool, and psoc
group_by(year, transect_name, psoc) |>
# calculate total percent cover across all quads
summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
# ungroup the data frame
ungroup() |>
# create a new column called year_pool from year and transect name
unite("year_pool", year, transect_name, remove = TRUE) |>
# join with metadata data frame
left_join(metadata, by = "year_pool") |>
# calculate mean percent cover from total percent cover/number of quadrats
mutate(mean_pc = sum_pc/num_quad)

# creating new object from veg
gl_native_explicit0 <- veg |>
# filter all occurrences where transect_name contains GL & native
filter(str_detect(transect_name, "GL") &
    site == "ncos" &
    cover_category == "NATIVE COVER") |>
# complete all possible combinations of year, pool, psoc and distance

```

```

complete(year, transect_name, psoc, transect_distance,
  # fill values of percent cover with 0
  fill = list(percent_cover = 0)
) |>
# group by year, pool, and psoc
group_by(year, transect_name, psoc) |>
# calculate total percent cover across all quads
summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
# ungroup the data frame
ungroup() |>
# create a new column called year_pool from year and transect name
unite("year_pool", year, transect_name, remove = TRUE) |>
# join with metadata data frame
left_join(metadata, by = "year_pool") |>
# calculate mean percent cover from total percent cover/number of quadrats
mutate(mean_pc = sum_pc/num_quad)

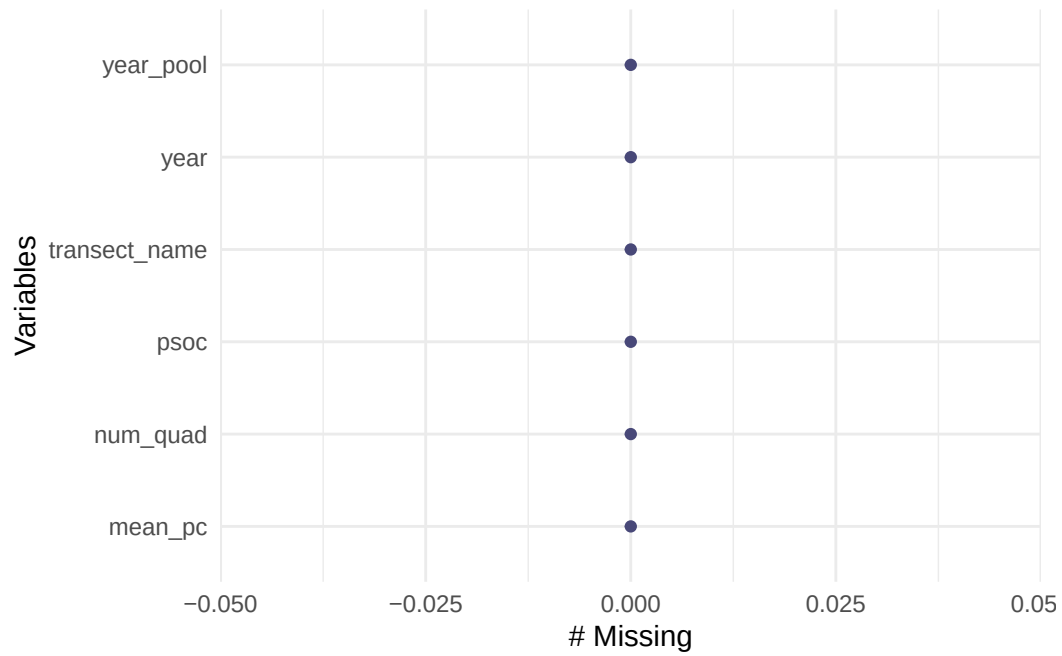
```

```

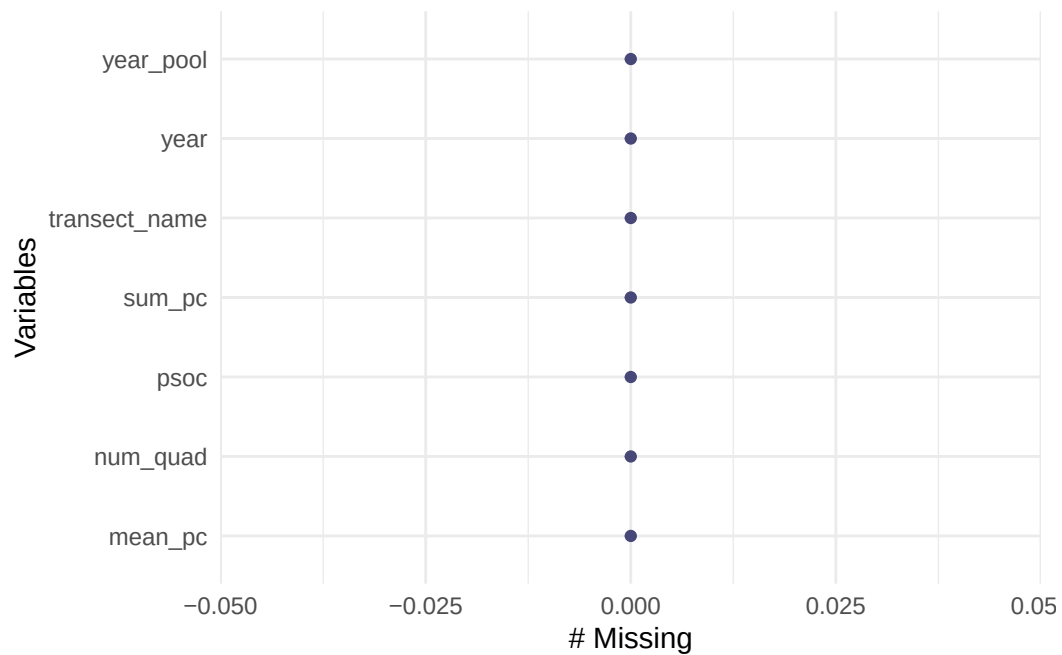
# creating new object from veg
gl_invasive_explicit0 <- veg |>
# filter all occurrences where transect_name contains GL & native
filter(str_detect(transect_name, "GL") &
  site == "ncos" &
  cover_category == "NON-NATIVE COVER") |>
# complete all possible combinations of year, pool, psoc and distance
complete(year, transect_name, psoc, transect_distance,
  # fill values of percent cover with 0
  fill = list(percent_cover = 0)
) |>
# group by year, pool, and psoc
group_by(year, transect_name, psoc) |>
# calculate total percent cover across all quads
summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
# ungroup the data frame
ungroup() |>
# create a new column called year_pool from year and transect name
unite("year_pool", year, transect_name, remove = TRUE) |>
# join with metadata data frame
left_join(metadata, by = "year_pool") |>
# calculate mean percent cover from total percent cover/number of quadrats
mutate(mean_pc = sum_pc/num_quad)

```

```
gg_miss_var(gl_implicitNA)
```



```
gg_miss_var(gl_explicit0)
```



No missing values in either data frame.

Grasslands vectors

```
# creating new vector from veg
native_spp <- veg |>
  # filter all occurrences where transect_name contains GL
  filter(str_detect(transect_name, "GL") &
    site == "ncos" &
    cover_category == "NATIVE COVER") |>
# pull to extract data frame column as a vector
pull(psoc)

# sample of ten rows
sample(native_spp, 10)
```

```
[1] "Heterotheca_grandiflora" "Deinandra_fasciculata"
[3] "Symphyotrichum_subulatum" "Stipa_pulchra"
[5] "Sidalcea_malviflora" "Epilobium_ciliatum"
[7] "Symphyotrichum_subulatum" "Juncus_phaeocephalus"
[9] "Eleocharis_acicularis" "Deinandra_fasciculata"
```

```
# creating new vector from veg
invasive_spp <- veg |>
  # filter all occurrences where transect_name contains GL and cover category is non-native
  filter(str_detect(transect_name, "GL") &
    site == "ncos" &
    cover_category == "NON-NATIVE COVER") |>
# pull to extract data frame column as a vector
pull(psoc)

# sample of ten rows
sample(invasive_spp, 10)
```

```
[1] "Brachypodium_distachyon" "Centaurium_tenuiflorum"
[3] "Hordeum_murinum" "Bromus_hordeaceus"
[5] "Polypogon_monspeliensis" "Bromus_diandrus"
[7] "Festuca_perennis" "Brachypodium_distachyon"
[9] "Festuca_perennis" "Festuca_myuros"
```

Species richness

```
# create new object from gl_explicit0
native_species_rich <- gl_explicit0 |>
# group by year and pool
  group_by(year, transect_name) |>
# filter data frame with vector into get native species,
# also filter percent cover to identify species presence
  filter(psoc %in% native_spp, mean_pc > 0) |>
# species richness calculation
  summarize(spp_rich = n_distinct(psoc)) |>
# ungroup data frame
  ungroup()

# sample 10 rows
slice_sample(native_species_rich, n = 10)
```

```
# A tibble: 10 x 3
   year transect_name spp_rich
  <dbl> <chr>         <int>
1  2022 GL-02             10
2  2024 GL-10             10
3  2022 GL-10              4
4  2025 GL-09              8
5  2022 GL-09              8
6  2024 GL-02             10
7  2024 GL-14             10
8  2023 GL-12              9
9  2024 GL-09              8
10 2022 GL-15              8
```

```
# create new object from gl_explicit0
invasive_species_rich <- gl_explicit0 |>
# group by year and pool
  group_by(year, transect_name) |>
# filter data frame with vector into get native species,
# also filter percent cover to identify species presence
  filter(psoc %in% invasive_spp, mean_pc > 0) |>
# species richness calculation
  summarize(spp_rich = n_distinct(psoc)) |>
# ungroup data frame
```

```

ungroup()

# sample 10 rows
slice_sample(invasive_species_rich, n = 10)

```

```

# A tibble: 10 x 3
  year transect_name spp_rich
  <dbl> <chr>         <int>
1  2022 GL-14          19
2  2024 GL-07          16
3  2022 GL-11          12
4  2023 GL-15          15
5  2024 GL-09          20
6  2022 GL-07          13
7  2025 GL-10          12
8  2025 GL-02          10
9  2021 GL-12          18
10 2025 GL-09          16

```

Outliers???

Total Cover

```

# create new object from veg
native_pc <- veg |>
  # filter only native species
  filter(psoc %in% native_spp) |>
  # filter only grasslands at NCOS
  filter(str_detect(transect_name, "GL") & site == "ncos") |>
  # group by year and transect
  group_by(year, transect_name) |>
  # calculate implicit missingness sum
  summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # united column to match the one in metadata
  unite("year_pool", year, transect_name, remove = TRUE) |>
  # join metadata

```

```

left_join(metadata, by = "year_pool") |>
# dividing total % cover by the # of quadrats in the observed year
mutate(total_pc = sum_pc/num_quad)

# sample of 10
slice_sample(native_pc, n = 10)

```

```

# A tibble: 10 x 6
  year_pool sum_pc year transect_name num_quad total_pc
  <chr>      <dbl> <dbl> <chr>          <dbl>      <dbl>
1 2025_GL-02    57 2025 GL-02          10        5.7
2 2022_GL-12   295 2022 GL-12          11       26.8
3 2025_GL-14   372 2025 GL-14          11       33.8
4 2021_GL-11   201 2021 GL-11          11       18.3
5 2024_GL-15   664 2024 GL-15          11       60.4
6 2021_GL-14   415 2021 GL-14          11       37.7
7 2024_GL-12   549 2024 GL-12          11       49.9
8 2022_GL-11   370 2022 GL-11          11       33.6
9 2025_GL-15   173 2025 GL-15          11       15.7
10 2025_GL-12   220 2025 GL-12          11        20

```

```

# create new object from veg
invasive_pc <- veg |>
# filter only invasive species
filter(psoc %in% invasive_spp) |>
# filter only grasslands at NCOS
filter(str_detect(transect_name, "GL") & site == "ncos") |>
# group by year and transect
group_by(year, transect_name) |>
# calculate implicit missingness sum
summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# united column to match the one in metadata
unite("year_pool", year, transect_name, remove = TRUE) |>
# join metadata
left_join(metadata, by = "year_pool") |>
# dividing total % cover by the # of quadrats in the observed year
mutate(total_pc = sum_pc/num_quad)

# sample of 10
slice_sample(invasive_pc, n = 10)

```



```
# A tibble: 10 x 6
  year_pool sum_pc year transect_name num_quad total_pc
  <chr>      <dbl> <dbl> <chr>                <dbl>      <dbl>
1 2022_GL-10 419 2022 GL-10                11      38.1
2 2025_GL-09 1022 2025 GL-09                11      92.9
3 2025_GL-15 682 2025 GL-15                11       62
4 2025_GL-14 1064 2025 GL-14                11     96.7
5 2023_GL-12 982 2023 GL-12                11     89.3
6 2023_GL-09 1485 2023 GL-09                11    135
7 2022_GL-14 894 2022 GL-14                11     81.3
8 2021_GL-11 266 2021 GL-11                11     24.2
9 2022_GL-12 364 2022 GL-12                11     33.1
10 2025_GL-11 768 2025 GL-11                11     69.8
```

Interest species wrangling

```
# native species
# species most significant: Stipa_pulchra and Distichlis_spicata
native_interest <- c(
  "Stipa_pulchra",
  "Distichlis_spicata",
  "Deinandra_fasciculata",
  "Erigeron_canadensis",
  "Symphyotrichum_subulatum"
)

# invasive / non-native species
# species most significant: Festuca_perennis and Polypogon_monspeliensis
non_native_interest <- c(
  "Festuca_perennis",
  "Polypogon_monspeliensis",
  "Bromus_hordeaceus",
  "Festuca_myuros",
  "Medicago_polymorpha"
)

# all selected species
all_interest <- c(
  "Stipa_pulchra",
  "Distichlis_spicata",
```

```

"Deinandra_fasciculata",
"Erigeron_canadensis",
"Symphyotrichum_subulatum",
"Festuca_perennis",
"Polypogon_monspeliensis",
"Bromus_hordeaceus",
"Festuca_myuros",
"Medicago_polymorpha"
)

```

```

gl_native_interest <- gl_native_explicit0 |>
  filter(psoc %in% native_interest) |>
  mutate(native = "native")

gl_invasive_interest <- gl_invasive_explicit0 |>
  filter(psoc %in% non_native_interest) |>
  mutate(native = "non_native")

gl_all_interest <- bind_rows(gl_native_interest, gl_invasive_interest)

```

```

# data frame for vis 1
gl_all_pc <- gl_all_interest |>
  group_by(psoc, year, native) |>
  summarize(sum_pc = sum(sum_pc),
            sum_quad = sum(num_quad)) |>
  ungroup() |>
  mutate(mean_pc = sum_pc/sum_quad)

```

```

# tbd

```

```

# data frames for vis 3
gl_native_tc <- gl_native_interest |>
  group_by(psoc, year_pool, native) |>
  summarize(sum_pc = sum(sum_pc)) |>
  ungroup()

gl_invasive_tc <- gl_invasive_interest |>
  group_by(psoc, year_pool, native) |>
  summarize(sum_pc = sum(sum_pc)) |>
  ungroup()

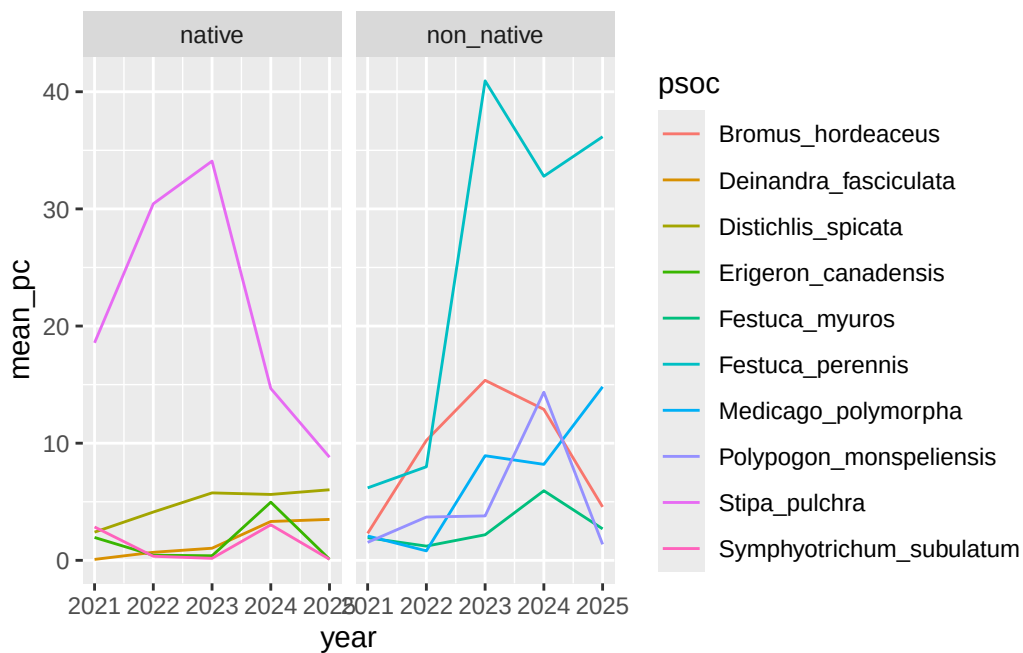
```

```
gl_stacked_native <- gl_all_interest |>
  group_by(native, year) |>
  summarize(mean_pc_sum = sum(mean_pc)) |>
  ungroup() |>
  mutate(native)
```

Visualizations

```
vis1 <- ggplot(data = gl_all_pc,
  aes(x = year,
    y = mean_pc,
    color = psoc)) +
  geom_line() +
  facet_wrap(~ native)
```

vis1



```
native_tc_plot <- ggplot(data = gl_native_tc,
  aes(x = psoc,
    y = sum_pc,
```

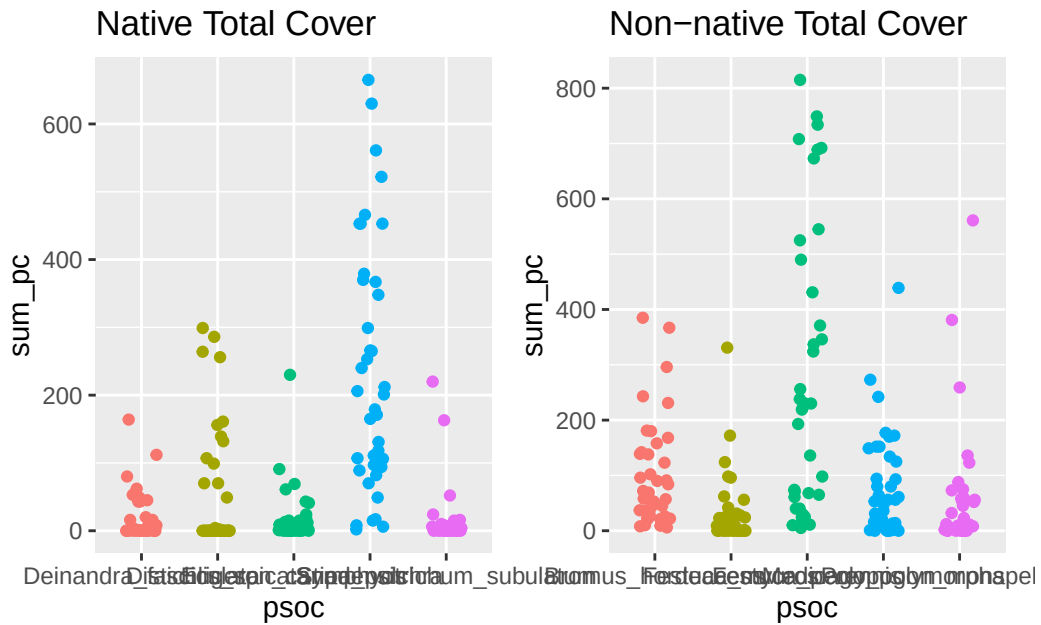
```

        color = psoc)) +
geom_jitter(height = 0,
            width = 0.2) +
stat_summary(geom = "pointrange",
            # argument to calculate standard error
            fun.data = mean_sdl,
            fun.args = list(mult = 1),
            size = 1,
            linewidth = 1,
            color = "red") +
labs(title = "Native Total Cover") +
theme(legend.position = "none")

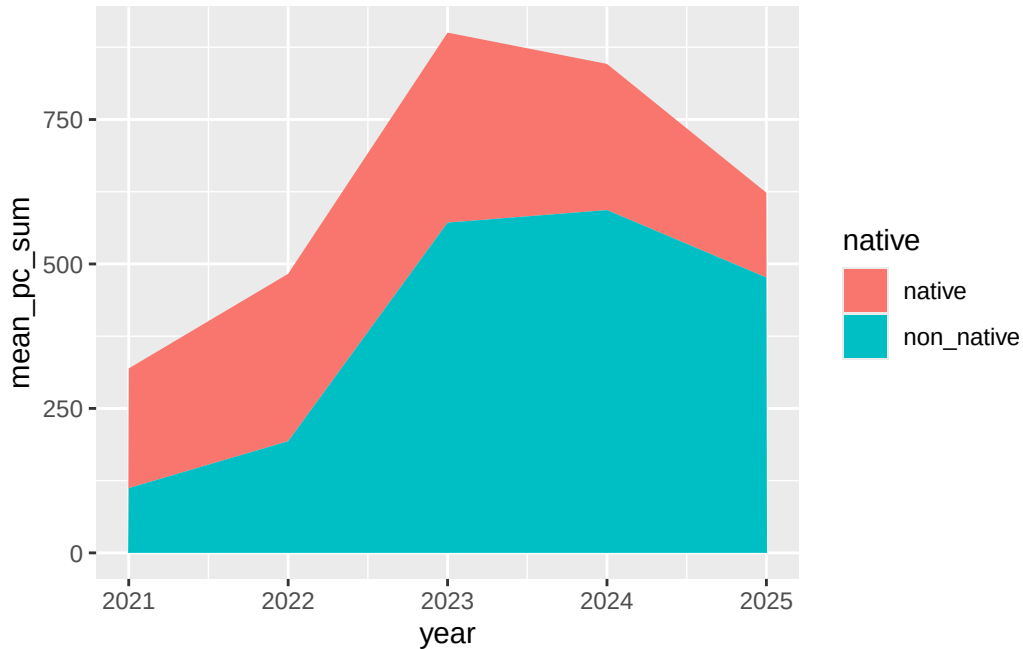
invasive_tc_plot <- ggplot(data = gl_invasive_tc,
    aes(x = psoc,
        y = sum_pc,
        color = psoc)) +
geom_jitter(height = 0,
            width = 0.2) +
stat_summary(geom = "pointrange",
            # argument to calculate standard error
            fun.data = mean_sdl,
            fun.args = list(mult = 1),
            size = 1,
            linewidth = 1,
            color = "red") +
labs(title = "Non-native Total Cover") +
theme(legend.position = "none")

vis3 <- native_tc_plot + invasive_tc_plot
vis3

```



```
vis4 <- ggplot(data = gl_stacked_native,
               aes(x = year,
                   y = mean_pc_sum,
                   fill = native)) +
  geom_area()
vis4
```



alyssa visualizations

non native and native pc over time FACETED by transect

```
gl_implicitNA_alyssa <- veg |>
  # filter all occurrences where transect_name contains VP
  filter(str_detect(transect_name, "GL") &
    site == "ncos") |>
  # group by year, pool, and species
  group_by(year, transect_name, psoc, cover_category) |>
  # calculate mean percent cover
  summarize(mean_pc = mean(percent_cover, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called year_pool from year and transect name
  unite("year_pool", year, transect_name, remove = TRUE) |>
  # join with metadata by year_pool
  left_join(metadata, by = "year_pool")

grass_cover <- gl_implicitNA_alyssa |>
```

```

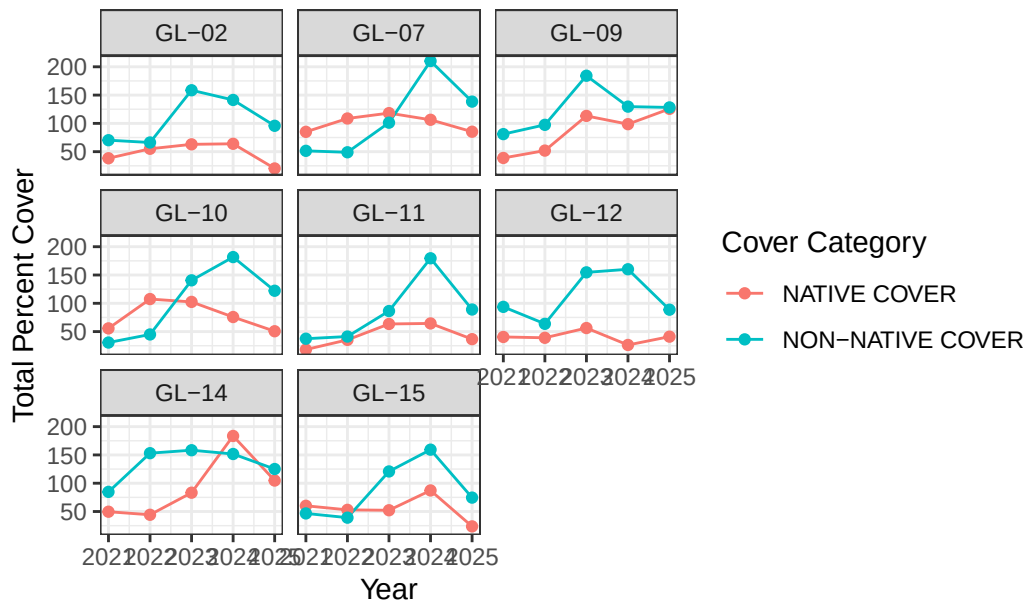
    filter(cover_category %in% c("NATIVE COVER", "NON-NATIVE COVER"))

gl_transects <- gl_implicitNA_alyssa |>
  filter(
    transect_name %in% c("GL-02", "GL-07", "GL-09", "GL-10",
                        "GL-11", "GL-12", "GL-14", "GL-15"),
    cover_category %in% c("NATIVE COVER", "NON-NATIVE COVER")
  )

gl_transects |>
  group_by(year, transect_name, cover_category) |>
  summarize(total_cover = sum(mean_pc, na.rm = TRUE)) |>
  ggplot(aes(x = year,
             y = total_cover,
             color = cover_category)) +
  geom_line() +
  geom_point() +
  facet_wrap(~ transect_name) +
  labs(
    title = "Native vs Non-Native Percent Cover Over Time Across GL Transects",
    x = "Year",
    y = "Total Percent Cover",
    color = "Cover Category"
  ) +
  theme_bw()

```

Native vs Non-Native Percent Cover Over Time Across GL Tra



non native and native species of interest over time FACETED by species

```
selected_species <- c(native_interest, non_native_interest)

selected_species_cover <- gl_implicitNA |>
  filter(psoc %in% selected_species) |>
  mutate(
    species_group = case_when(
      psoc %in% native_interest ~ "Native",
      psoc %in% non_native_interest ~ "Non-native"
    )
  )
```

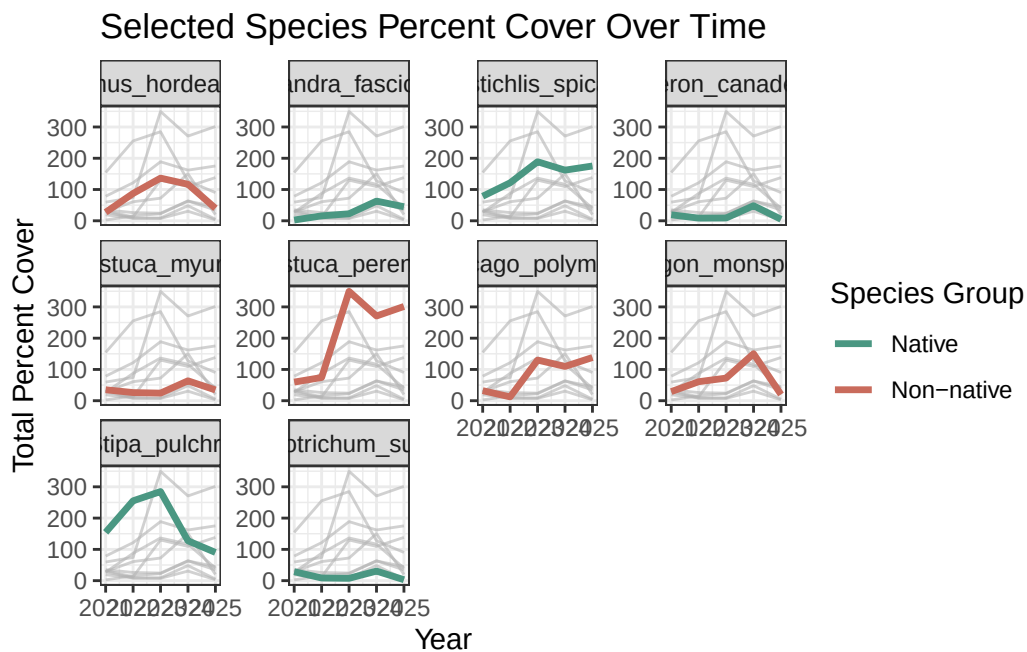
```
selected_species_cover |>
  group_by(year, psoc, species_group) |>
  summarize(total_cover = sum(mean_pc, na.rm = TRUE)) |>
  ggplot(aes(x = year,
             y = total_cover,
             color = species_group)) +
  geom_line(aes(group = psoc),
            linewidth = 1.2) +
```



```

gghighlight(
  use_direct_label = FALSE,
  unhighlighted_params = list(
    color = "gray70",
    alpha = 0.6,
    linewidth = 0.5
  )
) +
scale_color_manual(values = c(
  "Native" = "#4A9782",
  "Non-native" = "#C96B5C"
)) +
facet_wrap(~ psoc, scales = "free_y") +
labs(
  title = "Selected Species Percent Cover Over Time",
  x = "Year",
  y = "Total Percent Cover",
  color = "Species Group"
) +
theme_bw()

```



```

# create new object from gl_explicit0
# create new object from gl_implicitNA_alysa
alysa_species_richness <- gl_implicitNA_alysa |>
  # group by year, transect, and cover category
  group_by(year, transect_name, cover_category) |>
  # filter data frame to include only native and non-native species
  # also filter percent cover to identify species presence
  filter(cover_category %in% c("NATIVE COVER",
                              "NON-NATIVE COVER"),
         mean_pc > 0) |>
  # species richness calculation
  summarize(spp_rich = n_distinct(psoc)) |>
  # ungroup data frame
  ungroup()

```

bar plots

```

# combine native and non-native richness data
all_species_richness <- bind_rows(
  # add native label
  native_species_rich |>
    mutate(type = "Native"),
  # add non-native label
  invasive_species_rich |>
    mutate(type = "Non-native")
)

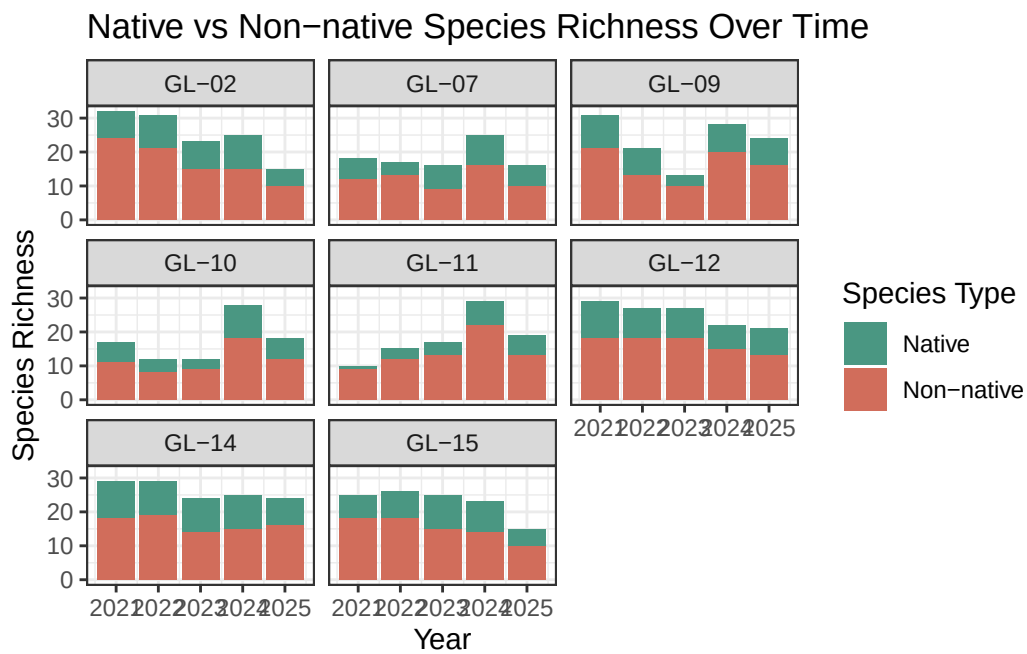
# create combined bar plot
species_richness_bar_plot <- all_species_richness |>
  # create ggplot
  ggplot(aes(x = year,
             y = spp_rich,
             fill = type)) +
  # create side-by-side bars
  geom_col() +
  # facet by transect
  facet_wrap(~ transect_name) +
  # customize colors
  scale_fill_manual(values = c(

```

```

  "Native" = "#4A9782",
  "Non-native" = "#D16D5B"
)) +
# add labels
labs(
  title = "Native vs Non-native Species Richness Over Time",
  x = "Year",
  y = "Species Richness",
  fill = "Species Type"
) +
# black and white theme
theme_bw()
# display plot
species_richness_bar_plot

```



```

# native species richness over time
native_species_rich_plot <- native_species_rich |>
# create ggplot
ggplot(aes(x = year,
            y = spp_rich,
            group = transect_name)) +
# create bars

```

```

geom_col(fill = "#4A9782") +
# facet by transect
facet_wrap(~ transect_name) +
# add labels
labs(
  title = "Native Species Richness Over Time",
  x = "Year",
  y = "Native Species Richness"
) +
# black and white theme
theme_bw()

```

```

# invasive/non-native species richness over time
invasive_species_rich_plot <- invasive_species_rich |>
# create ggplot
ggplot(aes(x = year,
           y = spp_rich,
           group = transect_name)) +
# create bars
geom_col(fill = "#D16D5B") +
# facet by transect
facet_wrap(~ transect_name) +
# add labels
labs(
  title = "Non-native Species Richness Over Time",
  x = "Year",
  y = "Non-native Species Richness"
) +
# black and white theme
theme_bw()

```

```

native_species_rich_plot / invasive_species_rich_plot

```

Native Species Richness Over Time



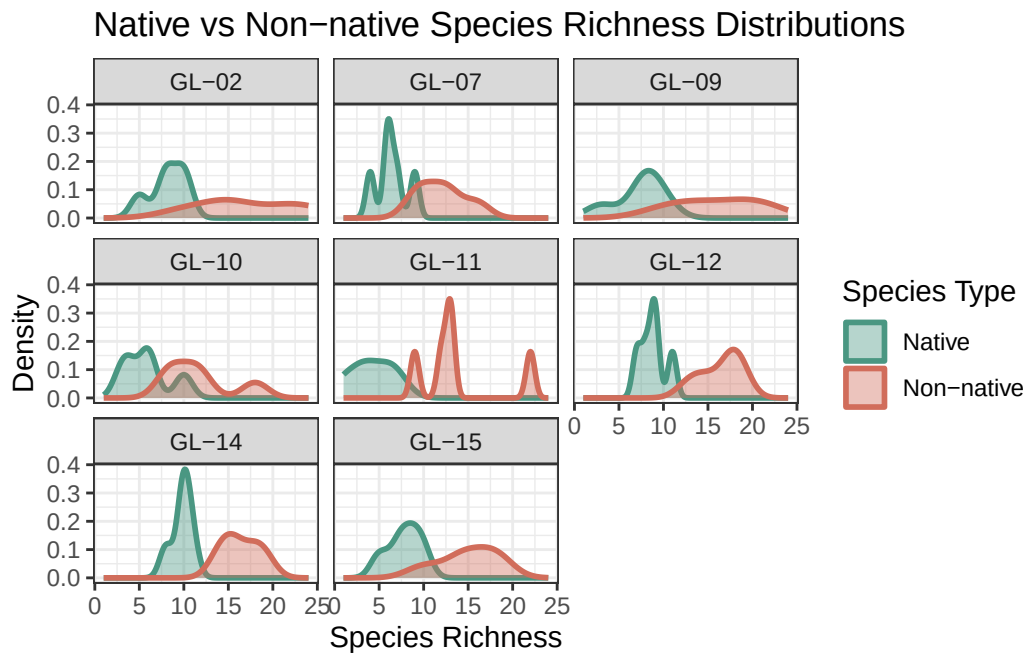
Non-native Species Richness Over Time



d plot all FINAL

```
# combine native and non-native richness data
all_species_richness <- bind_rows(
  # add native label
  native_species_rich |>
    mutate(type = "Native"),
  # add non-native label
  invasive_species_rich |>
    mutate(type = "Non-native")
)
# create combined density plot
species_richness_density_plot <- all_species_richness |>
  # create ggplot
  ggplot(aes(x = spp_rich,
             fill = type,
             color = type)) +
  # create overlapping density plots
  geom_density(alpha = 0.4,
              linewidth = 1) +
  # facet by transect
  facet_wrap(~ transect_name) +
  # customize colors
  scale_fill_manual(values = c(
    "Native" = "#4A9782",
    "Non-native" = "#D16D5B"
  )) +
  scale_color_manual(values = c(
    "Native" = "#4A9782",
    "Non-native" = "#D16D5B"
  )) +
  # add labels
  labs(
    title = "Native vs Non-native Species Richness Distributions",
    x = "Species Richness",
    y = "Density",
    fill = "Species Type",
    color = "Species Type"
  ) +
  # black and white theme
  theme_bw()
```

```
# display plot
species_richness_density_plot
```



REVISION - updated bar plot for all species

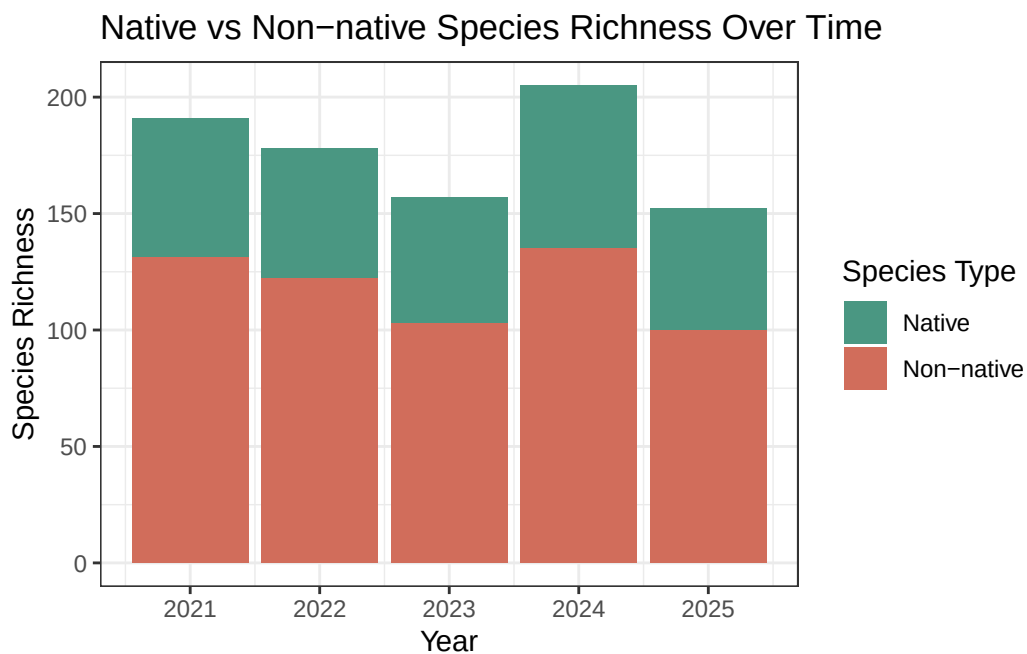
```
# combine native and non-native richness data
all_species_richness <- bind_rows(
  native_species_rich |>
    mutate(type = "Native"),
  invasive_species_rich |>
    mutate(type = "Non-native")
)

# summarize species richness by year and type
species_richness_summary <- all_species_richness |>
  group_by(year, type) |>
  summarize(total_richness = sum(spp_rich, na.rm = TRUE),
            .groups = "drop")

# create stacked bar plot over time
```

```
species_richness_bar_plot <- species_richness_summary |>
  ggplot(aes(x = year,
             y = total_richness,
             fill = type)) +
  geom_col() +
  scale_fill_manual(values = c(
    "Native" = "#4A9782",
    "Non-native" = "#D16D5B"
  )) +
  labs(
    title = "Native vs Non-native Species Richness Over Time",
    x = "Year",
    y = "Species Richness",
    fill = "Species Type"
  ) +
  theme_bw()

# display plot
species_richness_bar_plot
```



d plot by selected species richness

```
# create species richness object for selected species
selected_species_richness <- selected_species_cover |>
  # filter percent cover to identify species presence
  filter(mean_pc > 0) |>
  # group by year, transect, and native/non-native group
  group_by(year, transect_name, species_group) |>
  # count number of unique species present
  summarize(spp_rich = n_distinct(psoc),
    .groups = "drop")
```

REVISION - updated bar plot for selected species

```
# create species richness object for selected species
selected_species_richness <- selected_species_cover |>
  # filter percent cover to identify species presence
  filter(mean_pc > 0) |>
  # group by year, transect, and species group
  group_by(year, transect_name, species_group) |>
  # count unique species present
  summarize(spp_rich = n_distinct(psoc),
    .groups = "drop")
# summarize richness across years
selected_species_richness_summary <- selected_species_richness |>
  group_by(year, species_group) |>
  summarize(total_richness = sum(spp_rich, na.rm = TRUE),
    .groups = "drop")

# create stacked bar plot
selected_species_richness_bar_plot <- selected_species_richness_summary |>
  # create ggplot
  ggplot(aes(x = year,
    y = total_richness,
    fill = species_group)) +
  # create stacked bars
  geom_col() +
  # customize colors
  scale_fill_manual(values = c(
```

```

    "Native" = "#4A9782",
    "Non-native" = "#D16D5B"
  )) +
  # add labels
  labs(
    title = "Selected Species Richness Over Time",
    x = "Year",
    y = "Species Richness",
    fill = "Species Group"
  ) +
  # black and white theme
  theme_bw()
# display plot
selected_species_richness_bar_plot

```

